

1. A coupled partial melting Stokes-Darcy system. Strong form

$$\begin{aligned}
(1.1) \quad & \tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla (\phi_f^{-1/2} \tilde{q}_f) = 0 \\
& \mu_s \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0 \\
& \nabla q - \nabla \cdot \hat{\sigma}(\mathbf{v}_s) = -(1-\phi_f) \rho_r \mathbf{g} \\
& \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0
\end{aligned}$$

List SI units

q = pascal.

$\mathbf{v}$ = m/s

Non dimensionalization choice

- $\tilde{\mathbf{v}} = \frac{\tilde{\mathbf{v}}_r}{k_0 \rho_r g_y / \mu_f}$
- $\tilde{q} = \frac{q}{\rho_r g_y l_0}$
- $\tilde{x} = \frac{x}{l_0}$
- $l_0 = \left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2}$

The resulting non dimensionalized form is therefore

$$\begin{aligned}
(1.2) \quad & \tilde{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \tilde{q}_f) = 0 \\
& \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} \tilde{q}) = 0 \\
& \nabla \tilde{q} - \nabla \cdot \hat{\sigma}(\tilde{\mathbf{v}}_s) = -(1-\phi_f) \rho_r \hat{\mathbf{g}} \\
& \nabla \cdot \tilde{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} \tilde{q}) = 0
\end{aligned}$$

scaled nondimensionalized weak form let $\tilde{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3} \nabla \cdot \mathbf{v} \mathbf{I}$

$$\begin{aligned}
(1.3) \quad & (2\phi_s \tilde{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\tilde{q}_h, \nabla \cdot \Psi_s) = -(\phi_s \tilde{\mathbf{f}}, \Psi_s) \\
& (\nabla \cdot \tilde{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s} \tilde{q}_h, \omega \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \tilde{q}_{f,h}, \omega \right) = 0 \\
& (\tilde{\mathbf{v}}_{r,h}, \Psi_r) - (\tilde{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \Psi_r)) = 0 \\
& (\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \Psi_r), \omega_f) + \left(\frac{1}{\phi_s} \tilde{q}_{f,h}, \omega_f \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \tilde{q}_h, \omega_f \right) = 0
\end{aligned}$$

2. Linear system and Uzawa type algorithm. original linear system

$$(2.1) \quad \begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & -\mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & -\mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

This is a double saddle point system. Symmetrically permutate original linear system and form a enlarged saddle point system.

$$(2.2) \quad \begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & -\mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & -\mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$(2.3) \quad \tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_d \end{bmatrix} \quad \tilde{\mathbf{C}} = \begin{bmatrix} \mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & \mathbf{C}_d \end{bmatrix}$$

In this newly formed system, the $\tilde{\mathbf{A}}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [1], [2].

Consider the linear system of equation:

$$(2.4) \quad \begin{bmatrix} \mathbf{A} & -\mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix}$$

Algorithm 2.1 Inexact Uzawa Iteration

Initialize solution

2.1. Convergence and stability analysis.

2.2. Fixed point acceleration.

2.2.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [3]

3. Test cases. We test the linear system a little bit different from the linear system above.

3.1. Constant porosity and balanced pressure (divergence free velocity field).

3.2. Const porosity and unbalanced pressure.

$$(3.1) \quad \check{q} = 0 \quad \check{q}_f = 2(x + y) \quad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1 - \phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \quad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1 - \phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$$

$$(3.2) \quad \nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1 - \phi_f} (x + y)$$

$$(3.3) \quad \mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1 - \phi_f} \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \quad (\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I} = \frac{2\phi_f^{1/2}}{1 - \phi_f} \begin{bmatrix} x + y & 0 \\ 0 & x + y \end{bmatrix}$$

$$(3.4) \quad \begin{aligned} \check{\mathbf{v}}_r + \phi_f^{1/2} \nabla \check{q}_f &= -\frac{\phi_f^{-1/2}}{1 - \phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2} \nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1 - \phi_f} \check{q}_f &= \frac{-2}{1 - \phi_f} (x + y) + \frac{1}{1 - \phi_f} 2(x + y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1 - \phi_f} \check{q}_f &= \frac{2\phi_f^{1/2}}{1 - \phi_f} (x + y) - \frac{2\phi_f^{1/2}}{1 - \phi_f} (x + y) = 0 \\ -2(1 - \phi_f) \nabla \cdot \sigma(\check{\mathbf{v}}_s) &= -2(1 - \phi_f) \nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I}) = -\frac{8\phi_f^{1/2}}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

REFERENCES

- [1] H. C. Elman and G. H. Golub, Inexact and preconditioned uzawa algorithms for saddle point problems, *SIAM. J. Numer. Anal.*, 31 (1994), pp. 1645–1661.
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- [3] H. F. W. Nguyenho Ho, Sarah D. Olson, Accelerating the uzawa algorithm, *SIAM. J. Sci. Comput.*, 39 (2017), pp. S461–S476.